

1. (Fall 2024, #1) Evaluate the limits.

(a) $\lim_{x \rightarrow 0^+} (1 + x \ln(x))^{\frac{1}{x}}$

(b) $\lim_{x \rightarrow 0} \frac{\sin x - x \cos x - \sin(x^3/3)}{x^5}$

2. (Spring 2024, #1) Evaluate the following limits.

(a) $\lim_{x \rightarrow \infty} x \sinh\left(\frac{1}{x}\right)$

(b) $\lim_{x \rightarrow 0^+} x^{\sin x}$

3. (Fall 2023, #1) Evaluate the following limits.

(a) $\lim_{x \rightarrow 0^+} (\tan^{-1}(x))^{x^2}$

(b) $\lim_{x \rightarrow +\infty} \left(x \sin\left(\frac{1}{x}\right) - x \tan\left(\frac{1}{x}\right) \right)$

4. (Spring 2023, #1) Evaluate the following limits:

(a) $\lim_{x \rightarrow 0} \frac{2 \cos x + x^2 - 2}{x^4}$

(b) $\lim_{x \rightarrow +\infty} (x^3 e^{\frac{2}{x^3}} - x^3)$

5. (Fall 2022, #1) Evaluate the following limits:

(a) $\lim_{x \rightarrow 1} \frac{\ln x - x + 1}{(x - 1)^2}$

(b) $\lim_{x \rightarrow +\infty} x^{\frac{\ln x}{x}}$